# Aditya Kandukoori

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## EDUCATION

**Poolesville High School : SMCS Magnet Program** — **Poolesville, MD**

*Expected High School Diploma | Unweighted GPA: 3.93* — *August 2024 – June 2028*

**Completed Coursework:** Magnet Analysis (post-BC calculus), AP Calculus, AP Computer Science A, AP Biology, AP U.S. Government

**Junior Year Coursework (2026–27):** Linear Algebra, Discrete Mathematics, Applied Statistics, Introduction to Artificial Intelligence, AP Physics C (Mechanics & E&M), AP English Language, AP World History

## RESEARCH EXPERIENCE

**Algoverse AI Research Program** — **Remote**

*Student Researcher, PathoReason: Explanation Faithfulness in Pathology VLMs* — *May 2026 – Present*

- Co-authored a study testing whether pathology vision-language models' cited histological evidence actually drives their diagnoses, using a grid-referenced localization and occlusion protocol on the MHIST colorectal polyp benchmark.
- Designed and ran the perturbation experiments: in a pre-registered replication on 427 held-out tiles, masking the regions a model cites changed its diagnosis more often than masking control tissue (130 vs. 79 tiles, sign test $p = 5\times10^{-4}$).
- Found that stated features track the label: they matched the model's own diagnosis in 45 of 46 cases but expert consensus only at chance, and reversing explanation/classification order flipped 41 of 100 predictions in one direction with zero reverse flips (McNemar $p < 10^{-10}$), replicated on a second inference host.
- Built the full evaluation pipeline: grid-based evidence localization, biological-structure masking (stain deconvolution, nuclei segmentation), and statistics across 7,600+ model citations; no VLM tested passed a pre-registered competence bar.

**MoCo Innovation** — **Montgomery County, MD**

*Acting President* — *2024 – Present*

- Taught hands-on STEM classes to elementary students at Clarksburg Elementary School through a nonprofit serving underrepresented communities.
- Designed the lesson curriculum and instructional materials used in classroom delivery.

## PROJECTS

**AUMP: Machine Learning Early-Warning System for Bradycardic Deterioration**

*Sole Developer • PhysioNet Credentialed Researcher*

- Built a system predicting bradycardic deterioration in ICU patients from real clinical telemetry, using credentialed access to MIMIC-III and MIMIC-IV.
- Validated across two cohorts (up to 63,000+ patients) with verified patient-level train/test splits.
- Found an LSTM sequence model significantly outperforms a logistic regression baseline at every prediction horizon (patient-clustered bootstrap, 95% CIs excluding zero).
- Identified and corrected several methodological errors before finalizing results, including a label-timing bug and a patient-leakage risk.

**BUMP Labs : Wearable Closed-Loop Cardiac Response Device**

*Co-Founder • bump-labs.com • Patent Pending (USPTO provisional #63/686,176)*

Co-developed the original concept for a wearable atropine pump treating crashing bradycardia and contributed to the device codebase. Currently building an early-stage venture around the system, integrating the AUMP machine learning prediction layer with the BUMP hardware to create a closed-loop predict-and-treat platform.

**Comparative Analysis of Digital Tools and Traditional Teaching Methods**

*Published Research Paper • arXiv:2408.06689 • Second Author*

Co-authored a controlled study of 30 elementary students evaluating math skill gains from interactive digital platforms against worksheet-based instruction. Contributed to study design, assessment administration, and data analysis. Digital-tool students improved 24.2% on average versus 8.3% for the control group, demonstrating measurable gains from technology-assisted instruction.

**Falconwing : EMG-Controlled Robotic Prosthetic Arm**

Built a 3D-printed, Arduino-based prosthetic arm using MyoWare EMG sensors to translate real-time muscle signals into a controlled grasping mechanism.

## ACTIVITIES & SKILLS

- **Activities:** Taekwondo : Black Belt, Han Mi Martial Arts (Damascus, MD); Model United Nations : Delegate; 200+ volunteer service hours
- **Programming Languages:** Python, Java
- **ML & Data Science:** Scikit-learn, PyTorch, Pandas, NumPy, Hugging Face Transformers, Matplotlib
- **AI & Agentic Systems:** Vision-language model evaluation, LLM function calling & tool use, prompt engineering, causal perturbation analysis
- **Statistical Methods:** Time-series feature engineering, hypothesis testing (Wilcoxon, McNemar's, bootstrap resampling), patient-level cross-validation, ROC/AUPRC evaluation
- **Tools & Environments:** Git/GitHub, Jupyter Notebook, Kaggle, Google Colab, Arduino, PhysioNet/MIMIC